Hyperspectral Image Reconstruction From RGB Input Through Highlighting Intrinsic Properties

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***Abstract*— Dozens of spectral bands of hyperspectral images (HSIs) have been successfully reconstructed from only three color band images using deep neural networks according to their powerful nonlinear mapping capability. However, the existing deep-learning-based approaches tend to directly reconstruct HSIs from RGB inputs without emphasizing the discriminative intrinsic properties of different materials, resulting in certain distortion in reconstructed spectra. In this article, an intrinsic image decomposition (IID)-based spectral super-resolution (SSR) framework is proposed to reconstruct spectra of pixels from their reflectance feature and shading feature separately, by which the intrinsic properties can be emphasized during spectral reconstruction. Specifically, a dual hierarchical regression network (DHRNet) is designed for the proposed IID-based SSR task, in which a shading feature extraction module (SFEM) based on dense structure and a reflectance feature extraction module (RFEM) with attention mechanism are first, respectively, designed to reconstruct spectral information from reflectance feature and shading feature, and a feature enhancement module (FEM) is consequently devised to further improve the coarse combined estimation. Ultimately, a novel hybrid loss combining smooth l_1 loss, spectral angel mapper (SAM), and gradient prior is also presented to restrain the spectral distortion while enhancing the sharpness of the reconstructed HSI. Experimental results over three datasets demonstrate the superiority of our proposed framework.**

***Index Terms*— 3D-CNN, attention mechanism, intrinsic image decomposition (IID), multiscale learning, spectral super-resolution (SSR).**

I. Introduction

HYPERSPECTRAL remote sensing (RS) images [1] are characterized by high spectral resolution, which can reach the nanometer level, providing more abundant and accurate spectral information than the conventional images. Therefore, hyperspectral RS has been widely used in various fields, including agriculture [2], [3], [4], medical imaging [5], national security [6], [7], atmospheric environment [8], and even interstellar exploration. However, these advantages of hyperspectral RS come at a great cost. On the one hand, compared with RGB images, hyperspectral image (HSI) includes two spatial dimensions and one spectral dimension, which leads to a longer exposure time. On the other hand, due to the technical and economical limitations of electro-optical imaging sensors, most spectral imaging equipment has difficulty in obtaining images with high spatial and spectral resolutions simultaneously. The fact that the spatial resolution of HSI is usually much lower than that of its RGB counterpart seriously affects the development of hyperspectral imaging applications in various fields.

In recent years, with the continuous development of spectral imaging technology, plenty of methods have been proposed to balance the spectral and spatial resolutions in hyperspectral imaging. Among them, the super-resolution-based methods [9], [10], [11], [12], [13], [14], [15], [16], [17] are relatively effective approaches, which can be divided into two categories based on the input data: single-image-based methods, which directly restore HSI with high spatial and spectral resolutions from RGB image or multispectral image (MSI), and fusion-based methods [18], [19], which obtain HSI with high spatial and spectral resolutions by fusing high spatial resolution RGB image/MSI and low spatial resolution HSI. Tremendous works have demonstrated that the fusion-based methods are capable of generating better reconstructed HSI. However, obtaining properly registered RGB image/MSI and HSI simultaneously is extremely difficult in many practical scenarios, and the generalization of fusion-based methods still needs further investigation. Compared with HSI, RGB image can be obtained by consumer camera with lower cost much more easily. Meanwhile, there is a certain correlation between the pixel value of RGB image and the radiation value of HSI, which makes it possible to recover HSI from its RGB counterpart. Therefore, this article focuses on a single-image-based spectral super-resolution (SSR) method, particularly with RGB input, which can effectively reduce the cost of HSI acquisition.

In general, reconstructing HSI from RGB image is an extremely ill-posed process that attempts to establish a three-to-many mapping. Many methods [20], [21], [22], [23], [24] have been proposed to handle this inverse problem by using or designing various priors, including the sparsity and low rank of HSI. Sparse-coding methods use hyperspectral prior data to construct feature sparse dictionary, which is then mapped to obtain the RGB projected sparse dictionary and its corresponding coefficients; thus, target HSI can be reconstructed by the calculated coefficients. With the rapid development of deep learning, SSR methods using convolutional neural network (CNN) [25], [26], [27], [28], [29], [30], [31],

Manuscript received 6 May 2024; revised 3 July 2024; accepted 19 July 2024. Date of publication 1 August 2024; date of current version 12 August 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62171381 and Grant 62201445. *(Corresponding author: Shaohui Mei.)*

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Digital Object Identifier 10.1109/TGRS.2024.3436715

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[32] greatly improved hyperspectral computational imaging performance, compared with the traditional sparse recovery. Taking advantage of the powerful feature extraction capabilities of convolution operation, a considerable success has been achieved with CNN-based methods in SSR task. Despite of their impressive performance, the existing methods still suffer from several issues. On the one hand, current methods pay too much attention on designing complex networks to obtain better reconstruction results, while ignoring the utilization of distinguished features contained in RGB images, such that the crucial feature information cannot be fully learned and retained in the SSR process. On the other hand, the existing SSR methods do not fully consider the impact of environmental factors (such as lighting condition, noises, and other factors) on the reconstructed results, leading to poor SSR performance. Besides, current methods deal with spatial and spectral information in a relatively independent manner, lacking of consideration on constraining spatial–spectral consistency in the reconstruction process, which may further lead to structural inconsistency in the reconstructed HSI.

To tackle the abovementioned issues, an intrinsic image decomposition (IID)-based framework is proposed in this article. Our previous attempt has demonstrated that such IID-based framework is effective to reconstruct HSIs from RGB inputs [33]. In order to further improve the performance of IID-based framework, more complex and specific framework is designed to extract the decomposed IID features, and attention mechanism is also introduced to enhance the salient spatial and spectral features learning. Specifically, to retain more pure spectral and spatial features of original RGB inputs as much as possible, thus highlighting the intrinsic properties in hyperspectral domain, the IID is first introduced to decompose an RGB image into the reflectance and shading image. Notably, the reflectance image contains features only related to the material of object, which stands for the spectral features, while the shading image contains almost all external factors except for the reflection ability of object material itself for each band, referring to the spatial features. Hence, a novel dual hierarchical regression network (DHRNet) is designed to reconstruct the accurate representations of the two corresponding intrinsic features in hyperspectral domain, which is constructed in a dual-pathway structure with multiscale learning strategy. It is worth noting that the two decomposed intrinsic features are processed separately in DHRNet, where the shading feature is extracted by a shading feature reconstruction network (SFRN), while the reflectance feature is input to a reflectance feature reconstruction network (RFRN). After mapping the two intrinsic features into hyperspectral domain, a fusion operation is conducted to obtain a coarse estimation. To constrain the spatial–spectral consistency and improve the SSR performance, the intermediate result is then input to a feature enhancement module (FEM), where a reflectance attention mechanism block (RAMB) is proposed to further guide the spectral enhancement and 3-D CNN is introduced to make a spatial–spectral constraint. In a nutshell, the main contributions of this work are concluded as follows.

1) A novel IID-based framework is proposed to conduct SSR in a coarse-to-fine manner, in which HSIs are initially reconstructed from reflectance and shading components of RGB inputs using IID and then further fine-tuned through the proposed FEM to reduce spectral distortion. As a result, the intrinsic features, i.e., reflectance and shading features, can be emphasized during the spectral reconstruction. Specifically, a DHRNet is designed and trained in an end-to-end manner using a hybrid loss incorporated with smooth l_1 loss, spectral angel mapper (SAM), and gradient prior.
2) A reflectance feature extraction module (RFEM) is contrapuntally presented to process the reflectance counterpart through introducing attention mechanism, which is capable of refining the abundant spectral and spatial contextual information.
3) A shading feature extraction module (SFEM) is deployed with a residual dense structure based on the principle of IID that the brightness of RGB inputs at different wavelengths is thought to be fixed, which effectively extracts the local and global spatial information of shading counterpart.
4) An FEM is devised to refine the coarse IID-based estimation through employing the cross-attention mechanism and 3-D CNN, by which the spectral distortion can be further mitigated through jointly fine-tuning of spatial and spectral information.

The remaining sections of this article are organized as follows. Section II provides a summary of related work. Section III describes the proposed framework in detail. The experimental results on various datasets are analyzed and discussed in Section IV. Finally, conclusions are presented in Section V.

II. Related Work

In this section, a detailed review is presented for the study on SSR methods relevant to this work. Besides, a brief introduction of IID and its utilization in image processing is also introduced.

A. Spectral Super-Resolution

The early single-source RGB image-based SSR methods mostly adopt numerical regression methods to approximate the reconstructed HSI, such as the Wiener estimation method used in [20] and training-based regression analysis method [34]. These methods are computationally simple, but the results are far from satisfaction. In recent years, as single-source RGB image-based SSR technology attracts more and more attention, many novel methods have emerged.

The first sparse representation-based SSR algorithm was proposed in [21], in which the sparsity of natural spectral images was utilized, and an overcomplete dictionary was trained with prior spectral data. Then, the dictionary was reduced to RGB space using a color response function to solve sparse coefficients for SSR. However, the final SSR result was highly dependent on the quality of prior spectral dictionary. With the rapid development of machine learning technology in recent years, CNN has been widely used in SSR algorithms, which greatly improved hyperspectral

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computational imaging performance. Xiong et al. [35] proposed a unified end-to-end framework named HSCNN, which reconstructed the enhanced HSI from spectrally undersampled image. Shi et al. [36] further devised two advanced networks collectively named HSCNN+ based on the work of HSCNN, in which residual and dense strategies are introduced to obtain more excellent performance. Zhao et al. [37] proposed a hierarchical regression network (HRNet), which is able to fully explore the contextual information at different scales through PixelShuffle operations. Mei et al. [38] recently proposed a two-stage framework, by which an effective transformation model between RGB inputs and target HSIs is learned in a coarse-to-fine manner. Alvarez-Gila et al. [39] first applied generative adversarial networks (GANs) to RGB-to-HSI reconstruction. The method used cGANs as the basic network architecture and adopted a U-Net-based generator as the main component, establishing a joint spectral–spatial distribution generation model for natural HSIs to better utilize spatial contextual information and obtain more realistic HSIs. Inspired by the strong data regression ability of CNN architecture, Koundianya et al. [40] proposed 2-D–3-D CNN-based structures for reconstructing spectral data from RGB images. Recently, Li et al. [41] proposed a novel framework denoted as HSACS, which incorporated with deep hybrid 2-D–3-D CNN based on dual second-order attention with camera spectral sensitivity prior. The proposed HSACS is capable of excavating the spatial–spectral context information, thus obtaining considerable reconstruction performance. To solve the uneasy acquisition of pairwise HSI-RGB, Qu et al. [42] devised an unmixing guided unsupervised network (UnGUN), of which two unmixing branches and a reconstruction branch are incorporated to achieve the unsupervised SSR process without pairwise imagery.

However, all the existing methods apply RGB image to the framework as input directly without considering feature refinement of the RGB image, leading to a certain key feature loss and serious degradation in both spatial and spectral domains. Further improvement in SSR performance is highly expected by designing targeted SSR algorithm for different features of RGB input.

B. Intrinsic Image Decomposition

Barrow et al. [43] indicated that the scene could be decomposed into a set of intrinsic features, such as object surface reflectance, object geometry, lighting and shading, and scene depth. The process of recovering these intrinsic features from an image was defined as the IID.

In the early stage, the IID problem was modeled abstractly and then solved using various optimization methods, which made IID an ill-posed problem, and prior information was required for problem solving. The Retinex algorithm was a classic prior constraint method, which assumed that in the gradient map of original image, the reflectance feature corresponded to the region with larger gradients, while the shading part corresponded to the region with smaller gradient [44]. Therefore, the corresponding features can be obtained by minimizing the gradient maps of the shading and reflectance layers. With the development of deep learning, more and more effective approaches have been introduced to IID. Huang et al. [45] noticed that the spectral curves of the same surface may appear quite different due to the complex lighting and geometric structure of natural scenes. They proposed a low-rank IID (LRIID) model, which extended the perceptual model for RGB image intrinsic decomposition to multispectral domain, aiming to generate the shading and reflectance of a single multispectral image. A low-rank constraint was further introduced to reduce the ill-posedness of IID problem. Chen et al. [46] proposed a spectral IID (SIID) network, which was able to separate the intrinsic features of natural scenes, such as illumination, reflectance, and shading. SIID network was capable of dealing with more challenging tasks, such as separating the intrinsic images of scenes with position-variant geometry, providing assistance for many advanced computer vision tasks, including material classification and recognition, illumination shape, and spectral image super-resolution. It can be summarized that IID is now commonly regarded as separating an observed image into reflectance and shading feature maps. Reflectance feature is described as the ability of an object's surface to reflect incident light, which is an inherent property of the object material, independent of its surrounding environment and its own geometric shape. The shading feature includes object shadows, environmental highlights, and interobject reflections, which are the results of the interaction between object surface and lighting.

To this end, for the purpose of further performance improvement, a novel SSR framework based on IID is proposed to better deal with the discriminative characteristics of shading and reflectance feature in this work.

III. Proposed Method

A. IID of RGB Input

To fully exploit its features, the RGB image is first decomposed into a reflectance image and a shading image. Per the principle of IID, an image denoted as $\boldsymbol{I} \in \mathbb{R}^{W\times H\times C}$ with $\boldsymbol{W} \times \boldsymbol{H}$ pixels and $\boldsymbol{C}$ spectral bands can be modeled as the product of reflectance image $\boldsymbol{R} \in \mathbb{R}^{W\times H\times C}$ and shading image $\boldsymbol{S} \in \mathbb{R}^{W\times H\times C}$

$$\boldsymbol{I} = \boldsymbol{S} * \boldsymbol{R} \tag{1}$$

where $*$ represents the elementwise multiplication.

The introduced IID framework is based on an unsupervised architecture proposed by Lettry et al. [47], which is composed of an autoencoder with skip connections at each branch. As depicted in Fig. 1(b), the encoder and decoder deployed in each branch comprise of a succession of two convolution layers, which are all incorporated with a batch norm (BN) and ReLU operation. Specifically, the first convolution layer with 1×1 kernel size aims to project the input features into a space with 32-D. The second convolution is designed in 5×5 with 64 filters to extract the features effectively. Finally, an elementwise division is further conducted to enforce consistency of the decomposition. Based on the constraints that reflectance is fixed and shading is varying, a Siamese training process is implemented to train the network in Fig. 1(b), in which pairs of time-varying images from the same scene

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are required to generate the decomposition pair, and a joint loss, including the albedo similarity, shading chromaticity smoothness, and reconstruction consistency, is backpropagated to achieve the unsupervised process. Note that the inference phase of introduced IID framework only requires one single RGB image, which ensures the feasibility when conducting the IID of our datasets.

B. Reconstruction of Reflectance and Shading Feature

1) Overall Structure: In order to obtain the accurate representation of reflectance and shading feature in hyperspectral domain, a DHRNet is proposed, as shown in Fig. 1, which is designed in a dual-pathway structure, aiming to fully and separately exploit the decomposed features of RGB inputs. As depicted in Fig. 1(a), a shading feature reconstruction network (SFRN) is designed for shading feature reconstruction, and a reflectance feature reconstruction network (RFRN) is designed for reflectance feature reconstruction. Furthermore, a multiscale features extraction structure is employed in each reconstruction module for deeper feature mining, in which a PixelUnshuffle operation is employed to downsample the inputs and a PixelShuffle operation is employed to upsample the multiscale features after extraction. Let $\boldsymbol{S}_{\mathrm{RGB}} \in \mathbb{R}^{W\times H\times 3}$ and $\boldsymbol{R}_{\mathrm{RGB}} \in \mathbb{R}^{W\times H\times 3}$ denote the decomposed shading image and reflectance image, respectively; then, the inputs at the $\boldsymbol{i}$th scale in shading feature reconstruction path and reflectance feature reconstruction path of DHRNet can be represented as follows:

$$\boldsymbol{R}_i = \boldsymbol{DS}(\boldsymbol{R}_{\mathrm{RGB}}, \boldsymbol{r}) \tag{2}$$

$$\boldsymbol{S}_i = \boldsymbol{DS}(\boldsymbol{S}_{\mathrm{RGB}}, \boldsymbol{r}) \tag{3}$$

where $\boldsymbol{DS}(\cdot)$ stands for PixelUnshuffle operation and $\boldsymbol{r}$ is the ratio of downsampling.

As displayed in Fig. 1(a), SFEM is composed of a 2-D convolution layer and $\boldsymbol{N}_1$ stacked dense convolution attention blocks (DCABs), while RFEM is composed of a 2-D convolution layer and $\boldsymbol{N}_2$ residual dense blocks (RDBs) as well. Note that each module is followed by a 1×1 convolution layer defined as $\boldsymbol{F}_{C_{1\times 1}}$, which is deployed to adjust the channel number. Thus, the fully extracted deep feature information from RFEM and SFEM is denoted as $\boldsymbol{f}^i_{\mathrm{RFEM}}$ and $\boldsymbol{f}^i_{\mathrm{SFEM}}$, respectively, which can be described as follows:

$$\boldsymbol{f}^i_{\mathrm{SFEM}} = \boldsymbol{F}_{C_{1\times 1}}\left(\boldsymbol{F}^i_{\mathrm{RDB}_{N1}}\left(\ldots \boldsymbol{F}^i_{\mathrm{RDB}_1}\left(\boldsymbol{F}_{C_{3\times 3}}(\boldsymbol{S}_i)\right)\right) + \boldsymbol{S}_i\right) \tag{4}$$

$$\boldsymbol{f}^i_{\mathrm{RFEM}} = \boldsymbol{F}_{C_{1\times 1}}\left(\boldsymbol{F}^i_{\mathrm{DCAB}_{N2}}\left(\ldots \boldsymbol{F}^i_{\mathrm{DCAB}_1}\left(\boldsymbol{F}_{C_{3\times 3}}(\boldsymbol{R}_i)\right)\right) + \boldsymbol{R}_i\right) \tag{5}$$

where $\boldsymbol{F}_{C_{3\times 3}}$ represents the shallow feature extraction function operated by convolution with a kernel size of 3×3. $\boldsymbol{F}^i_{\mathrm{DCAB}_{N1}}$ represents the DCAB operation in the $\boldsymbol{i}$th branch, and $\boldsymbol{N}_1$ is the total number of DCAB stacked in the $\boldsymbol{i}$th branch in RFRN. Similarly, $\boldsymbol{F}^i_{\mathrm{RDB}_{N2}}$ denotes the RDB in the $\boldsymbol{i}$th branch, and $\boldsymbol{N}_2$ is the total number of RDB stacked in the $\boldsymbol{i}$th branch in SFRN. The final output of the $\boldsymbol{i}$th RFRN and SFRN branch is then represented as $\boldsymbol{f}^i_{\mathrm{Rout}}$ and $\boldsymbol{f}^i_{\mathrm{Sout}}$, which is able to be calculated through

$$\boldsymbol{f}^i_{\mathrm{Sout}} = \boldsymbol{F}_{\mathrm{cat}}\left(\boldsymbol{f}^i_{\mathrm{SFRN}}, \boldsymbol{US}\left(\boldsymbol{f}^{i+1}_{\mathrm{Sout}}, \boldsymbol{r}\right)\right) \tag{6}$$

$$\boldsymbol{f}^i_{\mathrm{Rout}} = \boldsymbol{F}_{\mathrm{cat}}\left(\boldsymbol{f}^i_{\mathrm{RFRN}}, \boldsymbol{US}\left(\boldsymbol{f}^{i+1}_{\mathrm{Rout}}, \boldsymbol{r}\right)\right) \tag{7}$$

where $\boldsymbol{F}_{\mathrm{cat}}$ is the function of concat operation. The extracted feature information of different branches is further upsampled by PixelShuffle operation denoted as $\boldsymbol{US}(\cdot)$ and gradually fused. With a convolution layer at size 1×1, the outputs of shading and reflectance feature in hyperspectral domain can be denoted as $\boldsymbol{S}_{\mathrm{HSI}} \in \mathbb{R}^{W\times H\times C}$ and $\boldsymbol{R}_{\mathrm{HSI}} \in \mathbb{R}^{W\times H\times C}$, which are actually equal to the output of the first branch belonged to each subnetwork.

2) RFEM: Reflectance feature is an intrinsic property of object, which contains abundant spectral and spatial information simultaneously. In order to focus on those channels and regions that are more meaningful to the SSR process, and thereby reduce spectral distortion, RFEM is composed of several DCABs, which are capable of extracting the local dense features associated with spectral and spatial dimension. Specifically, as shown in Fig. 1(c), the basic unit of DCAB consists of three convolution layers followed by ReLU activation function, where the present and previous block outputs are connected and transmitted to the next layer for local dense feature extraction. Then, the channel attention (CA) [48] and spatial attention (SA) [48] modules are incorporated to refine the important features.

3) SFEM: Shading feature only stands for the reflection ability of object material itself, which mainly contains spatial information. Thus, the extraction and reconstruction of spatial information are the main consideration when mapping the shading feature to hyperspectral domain. To effectively learn the local and global spatial information, RDB is considered as the main component of SFEM. As shown in Fig. 1(d), the adopted RDB is based on a residual dense structure, mainly composed of a four-layer convolution operation followed by a ReLU activation function.

C. Feature Enhancement Module

According to (1), a coarse target HSI $\boldsymbol{H}_{\mathrm{fused}}$ can be estimated through $\boldsymbol{S}_{\mathrm{HSI}} * \boldsymbol{R}_{\mathrm{HSI}}$. To bolster the performance in eliminating the spectral distortion while enhancing the spatial–spectral consistency, the FEM is proposed, as depicted in Fig. 1(e), where the cross-attention mechanism and 3-D CNN are introduced. Theoretically, reflectance feature can be regarded as the pure spectral information of objects. On the basis, an RAMB is devised, which incorporates the reflectance information into the final refining process for further spectral enhancement of $\boldsymbol{H}_{\mathrm{fused}}$. The two different inputs of RAMB $\boldsymbol{H}_{\mathrm{fused}}$ and $\boldsymbol{R}_{\mathrm{HSI}}$ are first reshaped into $(W \times H) \times C$. Subsequently, the reshaped $\boldsymbol{R}_{\mathrm{HSI}}$ is mapped into query $\boldsymbol{Q}$ and key $\boldsymbol{K}$, while the $\boldsymbol{H}_{\mathrm{fused}}$ is linearly projected into value $\boldsymbol{V}$. Afterward, the $\boldsymbol{Q}$ and $\boldsymbol{K}$ are multiplied and then processed through softmax to generate the importance weight of spectrum $\boldsymbol{R}_w$. Consequently, the $\boldsymbol{H}_{\mathrm{fused}}$ is able to be further guided through $\boldsymbol{R}_w$ along spectral dimension, thus enhancing the spectral reconstruction. The above process can be expressed as follows:

$$\boldsymbol{H}_{\mathrm{RAMB}} = \boldsymbol{V}\boldsymbol{R}_w \tag{8}$$

$$\boldsymbol{R}_w = \boldsymbol{F}_{\mathrm{softmax}}\left(\boldsymbol{K}^T\boldsymbol{Q}\right) \tag{9}$$

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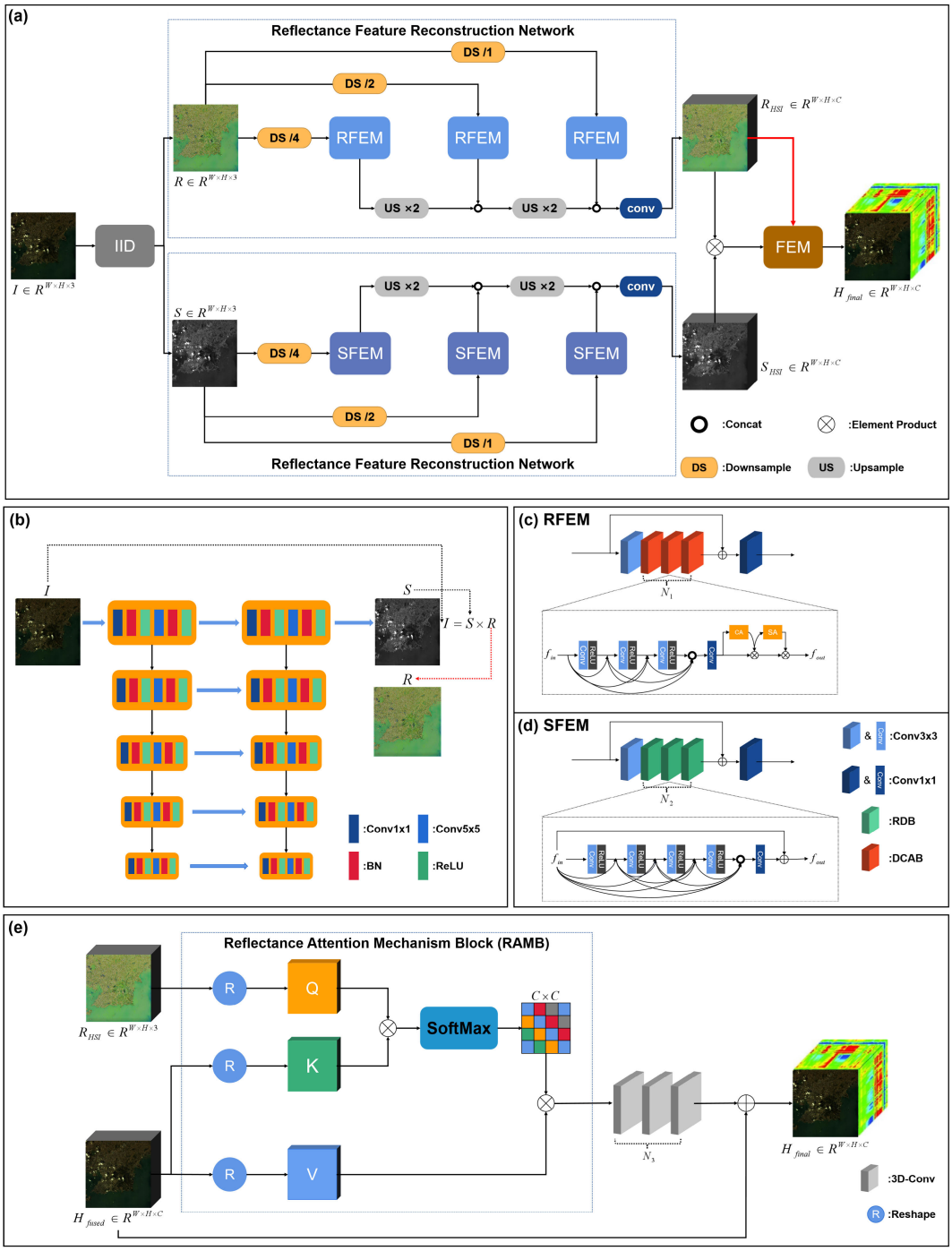


Fig. 1. Overall structure of the proposed DHRNet. (a) DHRNet. (b) IID. (c) RFEM. (d) SFEM. (e) FEM.

where $\boldsymbol{H}_{\mathrm{RAMB}}$ is the new weighted feature map in virtue of $\boldsymbol{R}_w$.

Considering that the existing SSR methods usually process spatial and spectral information in a relatively independent manner, which breaks the spectral consistency, result in the spectral bands to be disordered, $\boldsymbol{N}_3$ stacked 3-D CNN layers are further employed to enhance the final performance. As a result, the reconstructed HSI denoted as $\boldsymbol{H}_{\mathrm{final}}$ is calculated through

$$\boldsymbol{H}_{\mathrm{final}} = \boldsymbol{F}_{3D}^{N_3}\left(\ldots\left(\boldsymbol{F}_{3D}^{1}(\boldsymbol{H}_{\mathrm{RAMB}})\right)\right) + \boldsymbol{H}_{\mathrm{fused}}. \tag{10}$$

D. Loss Function

In the field of SSR, CNN-based methods often adopt MRAE or $\boldsymbol{l}_1$ as the loss function to measure numerical errors at pixel level. However, in order to obtain reconstructed HSI with accurate pixel values and high spectral fidelity, a hybrid constraint is designed, including a smooth $\boldsymbol{l}_1$ loss, a spectral fidelity loss, and a gradient constraint loss. Compared with $\boldsymbol{l}_1$, the introduced smooth $\boldsymbol{l}_1$ loss is more robust to the noise and differentiable at value 0, which is able to avoid the gradient explosion, thus obtaining higher reconstruction accuracy at pixel level. It can be defined

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as follows:

$$\boldsymbol{L}_{\text{smooth}_{l_1}} = \frac{1}{\boldsymbol{N}} \sum_{i=1}^{N} \begin{cases} 0.5(\boldsymbol{h}_i - \widetilde{\boldsymbol{h}}_i)^2, & \text{if } |\boldsymbol{h}_i - \widetilde{\boldsymbol{h}}_i| < 1 \\ |\boldsymbol{h}_i - \widetilde{\boldsymbol{h}}_i| - 0.5, & \text{otherwise} \end{cases} \tag{11}$$

where $\boldsymbol{h}_i$ and $\widetilde{\boldsymbol{h}}_i$ represent the $\boldsymbol{i}$th pixel in the reconstructed HSI and its corresponding GT. Note that $\boldsymbol{N}$ stands for the total number of pixels contained in an HSI, which equals $\boldsymbol{W} \times \boldsymbol{H} \times \boldsymbol{C}$.

Considering the abundant spatial and spectral information contained in HSI, it should not be only measured by the difference between corresponding pixel values to ensure the spatial fidelity, but also be evaluated for spectral dimension to reduce spectral degradation. Therefore, a spectral fidelity loss based on SAM is designed to constrain the distance between the generated HSI and target HSI in spectral domain

$$\boldsymbol{L}_{\text{SAM}} = \frac{1}{\boldsymbol{n}} \sum_{i=1}^{n} \left(\arccos \left(\frac{\boldsymbol{H}_i \cdot \widetilde{\boldsymbol{H}}_i}{\|\boldsymbol{H}_i\|_2 \times \left\|\widetilde{\boldsymbol{H}}_i\right\|_2} \right) \right) \tag{12}$$

where $\boldsymbol{H}_i$ and $\widetilde{\boldsymbol{H}}_i$ denote the spectral vector at the $\boldsymbol{i}$th pixel and $\boldsymbol{n}$ stands for the total number of pixels belonged to each band, which equals $\boldsymbol{W} \times \boldsymbol{H}$. Meanwhile, a gradient constraint is designed to provide additional supervision for better image reconstruction, aiming to ensure the structural consistency. The gradient loss is formulated by diminishing the distance between the gradient map extracted from the reconstructed HSI and the corresponding ground truth as follows:

$$\boldsymbol{L}_{\text{GRA}} = \frac{1}{\boldsymbol{N}} \sum_{i=1}^{N} \left\| \boldsymbol{G}(\boldsymbol{h}_i) - \boldsymbol{G}(\widetilde{\boldsymbol{h}}_i) \right\|_1 \tag{13}$$

where $\boldsymbol{G}(\cdot)$ stands for the operation of gradient map extraction.

In summary, the final hybrid loss is then defined as a weighted combination of the abovementioned losses

$$\boldsymbol{L}_{\text{hybrid}} = \boldsymbol{L}_{\text{smooth}_{l_1}} + \alpha \boldsymbol{L}_{\text{SAM}} + \beta \boldsymbol{L}_{\text{GRA}} \tag{14}$$

where α and β are two hyperparameters to balance the proportion of various losses in the final hybrid loss.

E. Network Configuration

The proposed DHRNet is based on a dual-pathway multiscale architecture, which aims to fully explore shading and reflectance features decomposed from RGB input. The number of branches in each path is set to 3, in which the maximum downsampling input is 1/4 of the original spatial resolution. Considering the discriminative features, the deployment strategy of different base units contained in SFEM and RFEM is distinguishable, where the number of N_1 and N_2 is set to 2, 4, 5 and 1, 2, 3 in different scale branches, respectively. As for the proposed FEM, the number of N_3 is set to 4, which are stacked sequentially with a residual structure. Note that batch normalization in DHRNet is removed to avoid the limitation of learning partial information. Two hyperparameters in loss function, α and β, are both set to 0.5.

IV. Experimental Evaluation

A. Datasets and Experimental Settings

Three datasets are adopted for evaluation, including two RS datasets and one natural image dataset.

1) RS Dataset: As depicted in Fig. 2, the first RS dataset is an available collection from the NASA Jet Propulsion Laboratory (JPL) gathered by AVIRIS sensor, denoted as Cuprite, which contains five scenes, covering valleys, rivers, mountains, and forests. Among these scenes, the first four ones are at a spatial size of 512 × 614, and the fifth one is at 158 × 614. Each scenario ranges from 369.85 to 2506.81 nm with 224 bands. Notably, a visible range (409–769 nm with 41 bands) is chosen from the original dataset to form the target HSIs. Another RS dataset is collected by Chinese Gaofen-5 satellite images, named GF5, which has a variety of objects, including urbans, mountains, harbors, and so on. As described in Fig. 3, ten HSIs are chosen in this work, which has a spatial size of 2008 × 2003 with 150 bands ranging from 390 to 1035 nm. To better evaluate our proposed model, 31 bands in the visible range with 10-nm spectral resolution of original ten HSIs are selected. In Cuprite dataset, three scenes are selected as the training data, and the rest two images are treated as the validation data, as illustrated in Fig. 2. Besides, as highlighted in Fig. 3, eight images in GF5 dataset are randomly selected as the training set, and the rest two serve as the validation set.

2) Natural Image Dataset: One benchmark natural image dataset is further introduced to make a comparison, that is, the CAVE dataset acquired by spectral camera of "Apogee Alta U260" by Columbia University [49]. The CAVE dataset contains 31 scenes with many discriminative indoor objects, including fruits, paintings, faces, and so on, which covers a spectral range from 400 to 700 nm at 10-nm increments with 31 bands and 512 × 512 pixels in each spectral band. For the CAVE dataset, 24 paired scenes are randomly chosen for training, and the remaining images are regarded as the validation set.

Specially, to generate the corresponding RGB counterpart of HSI, a fixed response function belonged to each dataset is applied to the corresponding reference HSI, which can be defined as follows:

$$\boldsymbol{I}_{\text{RGB}} = \boldsymbol{I}_{\text{HSI}} \times \boldsymbol{F}_{\text{response}} \tag{15}$$

where $\boldsymbol{I}_{\text{RGB}} \in \mathbb{R}^{H \times W \times 3}$ represents the generated RGB image, $\boldsymbol{I}_{\text{HSI}} \in \mathbb{R}^{H \times W \times C}$ stands for the ground-truth HSI cube, and $\boldsymbol{F}_{\text{response}} \in \mathbb{R}^{3 \times C}$ is the response function ($\boldsymbol{C}$ is set to 31 and 41 according to the datasets in this work). Note that GF5 dataset has provided paired HSI and RGB images, while the corresponding RGB images of other abovementioned datasets are obtained by the fixed response function.

3) Implementation Details: To eliminate the influence of initialized weights, a deterministic random seed is employed to ensure that the initialized weights of the proposed DHRNet are the same for each training; that is, the DHRNet is initialized with orthogonal initialization. In training, both RGB images and HSIs are linearly rescaled to [0, 1], and sample pairs with 128 × 128 pixels are cropped from the dataset with a stride at 8 pixels. Specifically, the RGB image is rotated 90°/180°/270° and flipped up/down/left/right to further augment training dataset. A minibatch size of 20 and Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$ are empirically selected. The initial learning rate is set as 0.001 and is decayed by ten times after 300 epochs, while the total epoch is 500. The proposed model

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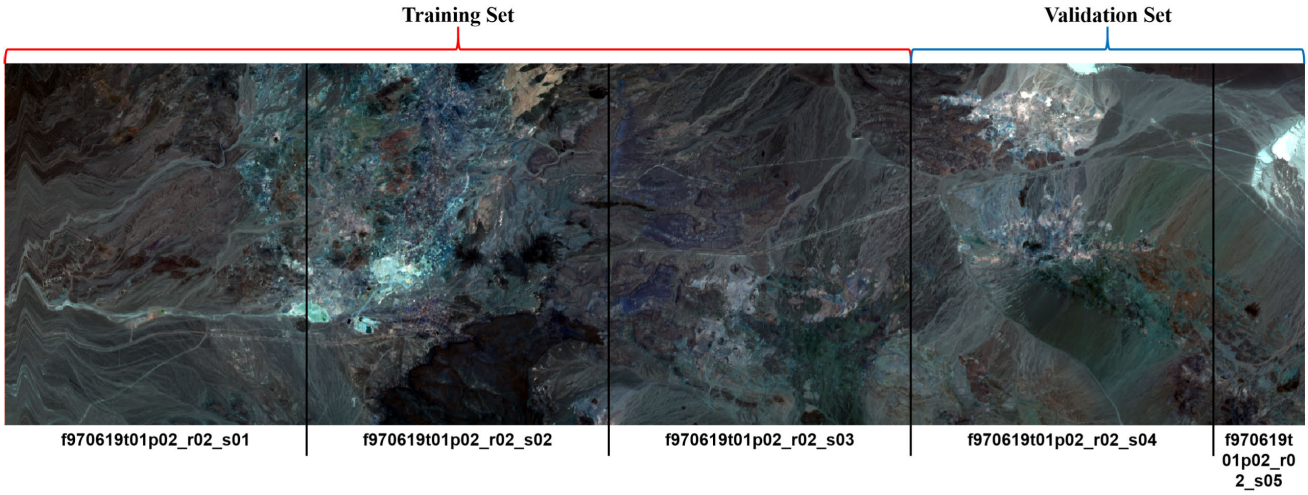


Fig. 2. Scenes of RS images in the Cuprite dataset are displayed in pseudocolor.

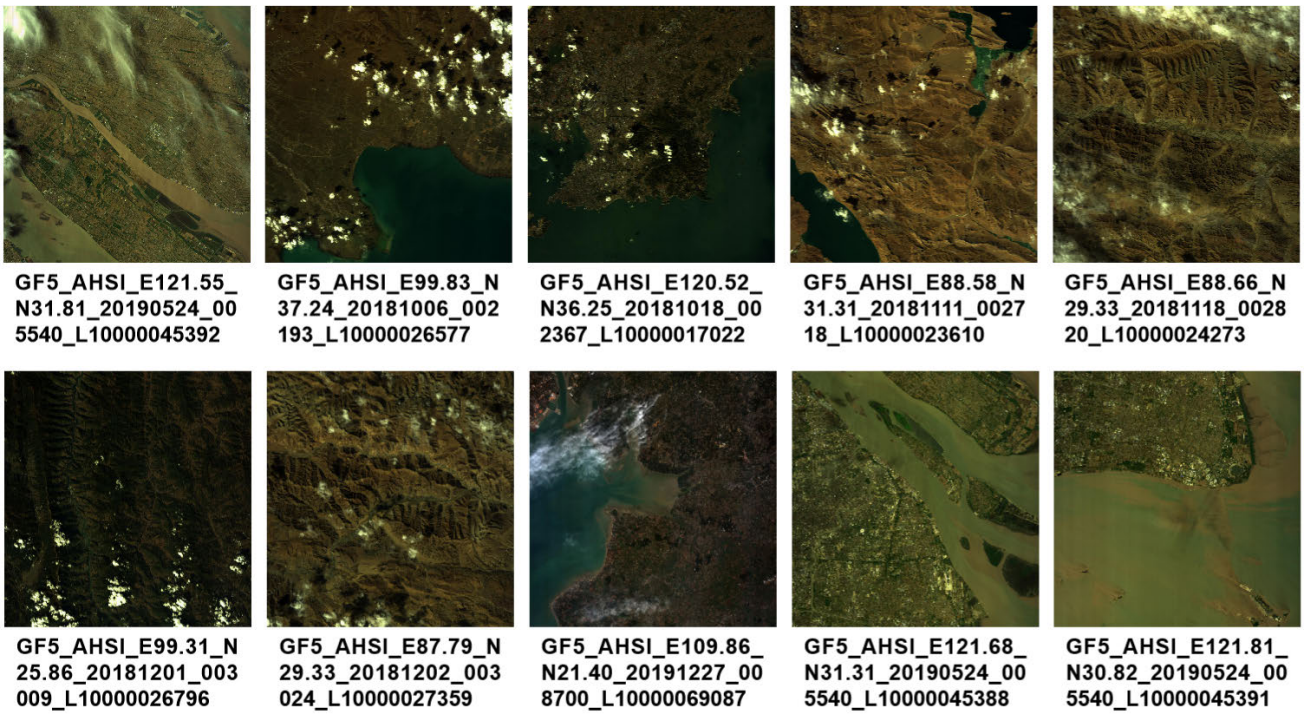


Fig. 3. Scenes of RS images in the GF5 dataset are displayed in RGB.

and experiments are implemented by PyTorch on NVIDIA GTX 2080Ti.

4) Evaluation Metrics: The evaluation standard follows [16] and [26], in which root-mean-square error (RMSE), peak signal-to-noise ratio (PSNR), structural similarity (SSIM), and SAM are considered as the quantitative metrics. RMSE measures the pixelwise square root error between the generated and real spectra, which is defined as follows:

$$\mathbf{RMSE}(\boldsymbol{h}, \boldsymbol{H}) = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{h}_i - \boldsymbol{H}_i)^2} \tag{16}$$

where $\boldsymbol{h}$ and $\boldsymbol{H}$ are two scenes and $\boldsymbol{N}$ is the total number of pixels contained in $\boldsymbol{h}$ and $\boldsymbol{H}$. Besides, PSNR is also included in this article, which is often used as a measurement of signal reconstruction quality in image compression and other fields. PSNR is mainly used to evaluate the reconstruction performance from the perspective of spatial distribution, which reflects the relative magnitude between the reconstructed and real HSIs

$$\mathbf{MSE}(\boldsymbol{h}, \boldsymbol{H}) = \frac{1}{N}\sum_{i=1}^{N}(\boldsymbol{h}_i - \boldsymbol{H}_i)^2 \tag{17}$$

$$\mathbf{PSNR}(\boldsymbol{h}, \boldsymbol{H}) = 20 \times \lg\left(\frac{\mathbf{MAX}}{\sqrt{\mathbf{MSE}}}\right) \tag{18}$$

where **MAX** stands for the maximum pixel value. SSIM is used to measure the spatial structure similarity between two images and, furthermore, to measure the compatibility of the reconstructed results with human visual system, which is

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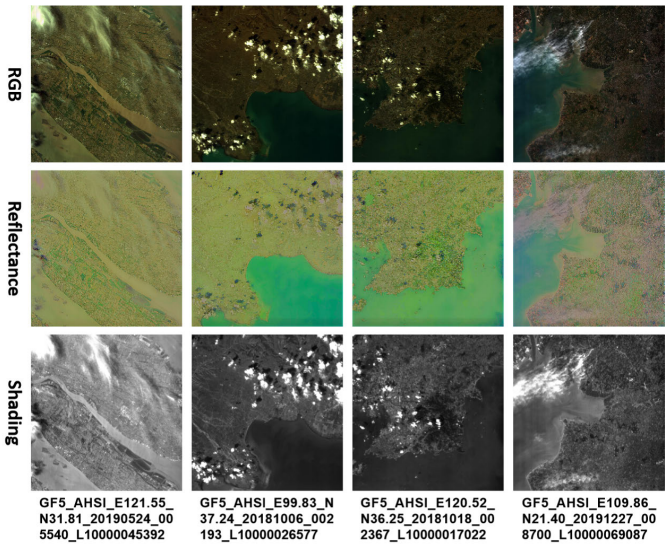


Fig. 4. IID results from different datasets. Best viewed on the screen.

TABLE I
QUANTITATIVE INDICATORS OF DIFFERENT METHODS ON RS DATASETS

Datasets	Method	RMSE↓	PSNR↑	SSIM↑	SAM↓
Cuprite	HSCNN-R [36]	0.0085	42.14	0.9945	1.58
	HRNet [37]	0.0070	44.41	0.9954	1.73
	EDSR [32]	0.0069	44.15	0.9953	1.68
	HDNet [30]	0.0072	44.19	0.9961	1.70
	MIRNet [50]	0.0066	44.76	0.9964	1.49
	HSACS [41]	0.0065	44.85	0.9969	1.50
	DHRNet	**0.0061**	**45.20**	**0.9973**	**1.47**
GF5	HSCNN-R [36]	0.0710	24.09	0.8192	3.26
	HRNet [37]	0.0512	26.43	0.8799	3.02
	EDSR [32]	0.0560	25.64	0.8944	2.54
	HDNet [30]	0.0511	26.57	0.9378	2.52
	MIRNet [50]	0.0529	26.25	0.9372	2.78
	HSACS [41]	0.0420	29.17	0.9224	2.83
	DHRNet	**0.0383**	**29.84**	**0.9395**	**2.48**

presented as follows:

$$\mathbf{SSIM}(\boldsymbol{h}, \boldsymbol{H})_k = \frac{(2\boldsymbol{\mu}_h\boldsymbol{\mu}_H + \boldsymbol{c}_1)(2\boldsymbol{\sigma}_{hH} + \boldsymbol{c}_2)}{(\boldsymbol{\mu}_h{}^2 + \boldsymbol{\mu}_H{}^2 + \boldsymbol{c}_1)(\boldsymbol{\sigma}_h{}^2 + \boldsymbol{\sigma}_H{}^2 + \boldsymbol{c}_2)} \quad (19)$$

where $\boldsymbol{\mu}_h$, $\boldsymbol{\mu}_H$, $\boldsymbol{\sigma}_h$, and $\boldsymbol{\sigma}_H$ represent the means and variances of $\boldsymbol{h}$ and $\boldsymbol{H}$, respectively, and $\boldsymbol{\sigma}_{hH}$ is the covariance of two images. Besides, $\boldsymbol{c}_1$ and $\boldsymbol{c}_2$ are two manual constants, which are related to the range of pixel value. SAM is defined to measure the spectral angle distance between two images along the spectral dimension, indicating the accuracy of reconstructed spectrum, whose detailed definition has been illustrated in (12).

B. Experimental Comparisons and Analysis

To illustrate the superiority of the proposed framework, six related methods are employed for comparison, including the HSCNN-R [36], HRNet [37], EDSR [32], HDNet [30], MIRNet [50], and HSACS [41]. Among these competitors, HSCNN-R and HRNet can be regarded as the most representative deep-learning-based SSR framework, which are based on the classical dense and multiscale learning strategy. Besides, EDSR, HDNet, and MIRNet are three natural image restoration models, which has been proved to be very effective in SSR task. HSACS is one latest SSR framework, which introduces the camera spectral sensitivity as the prior information and develops a deep dual second-order attention-based hybrid 2-D–3-D CNN framework to acquire SOTA results.

1) Results on RS Datasets: Table I illustrates the quantitative indices on Cuprite and GF5 datasets, in which the best ones are in bold and the suboptimal ones are underlined. It is obvious that our proposed DHRNet clearly outperforms the compared methods in terms of all the indicators. Specifically, compared with the SOTA method HSACS, our proposed framework improves the PSNR by 0.67 and 0.35 dB in GF5 dataset and Cuprite dataset, respectively. Meanwhile, DHRNet also makes an improvement in SSIM, which is clearly increased by 0.0004 and 0.0017 compared with the second-best results in two datasets. Notably, the superior amelioration is in RMSE and SAM. Compared with HSACS, DHRNet achieves better results on RMSE by reducing about 8.81% and 6.15% on GF5 and Cuprite datasets, respectively. Besides, in comparison with our framework, the suboptimal SAM achieved by MIRNet and HDNet still lags behind 1.34% and 1.59% on two RS datasets.

Visual results are also provided to further verify the effectiveness of our framework. IID results (reflectance image and shading image) of four selected scenes from GF5 are depicted in Fig. 4. It can be observed that the reflectance image contains abundant color and texture information, whereas the information contained in shading image is relatively simpler. Figs. 5 and 6 further compare the absolute error heat maps of different methods at various bands (GF5 dataset is displayed at 424, 501, 578, 595, 621, and 647 nm, and Cuprite dataset is demonstrated at 409, 438, 673, 692, 712, and 731 nm), in which the columns present the difference maps between the randomly selected bands reconstructed from seven different SSR models (six competitive methods and the proposed DHRNet). Besides, the selected bands of GT are also displayed at the last column of visual results. Note that the absolute error map represents the reconstruction accuracy in spatial domain, where darker color indicates better results. These visual results demonstrate that DHRNet is capable of reconstructing HSI more similar to the GT both in structure and texture. Dramatically, in Fig. 5, most of the methods can recover satisfactory results at 409 and 438 nm. However, the performance of other methods decreases obviously with the increase of wavelength, and DHRNet still has prominent reconstructed results.

In addition, spectral curves are further described in Figs. 7 and 8, aiming to illustrate the reconstructed results in spectral domain. The spectral curves of four randomly chosen points from different SSR methods are compared with GT in Fig. 7, demonstrating that the spectral curves in HSIs reconstructed by the proposed DHRNet are much more similar to the reference ones. Fig. 8 consists of two subgraphs, where (a) and (b) are the spectral curves of different pixels from DHRNet and GT. In Fig. 8, solid lines represent the GT, and dashed lines denote the reconstructed results, from which it can be observed

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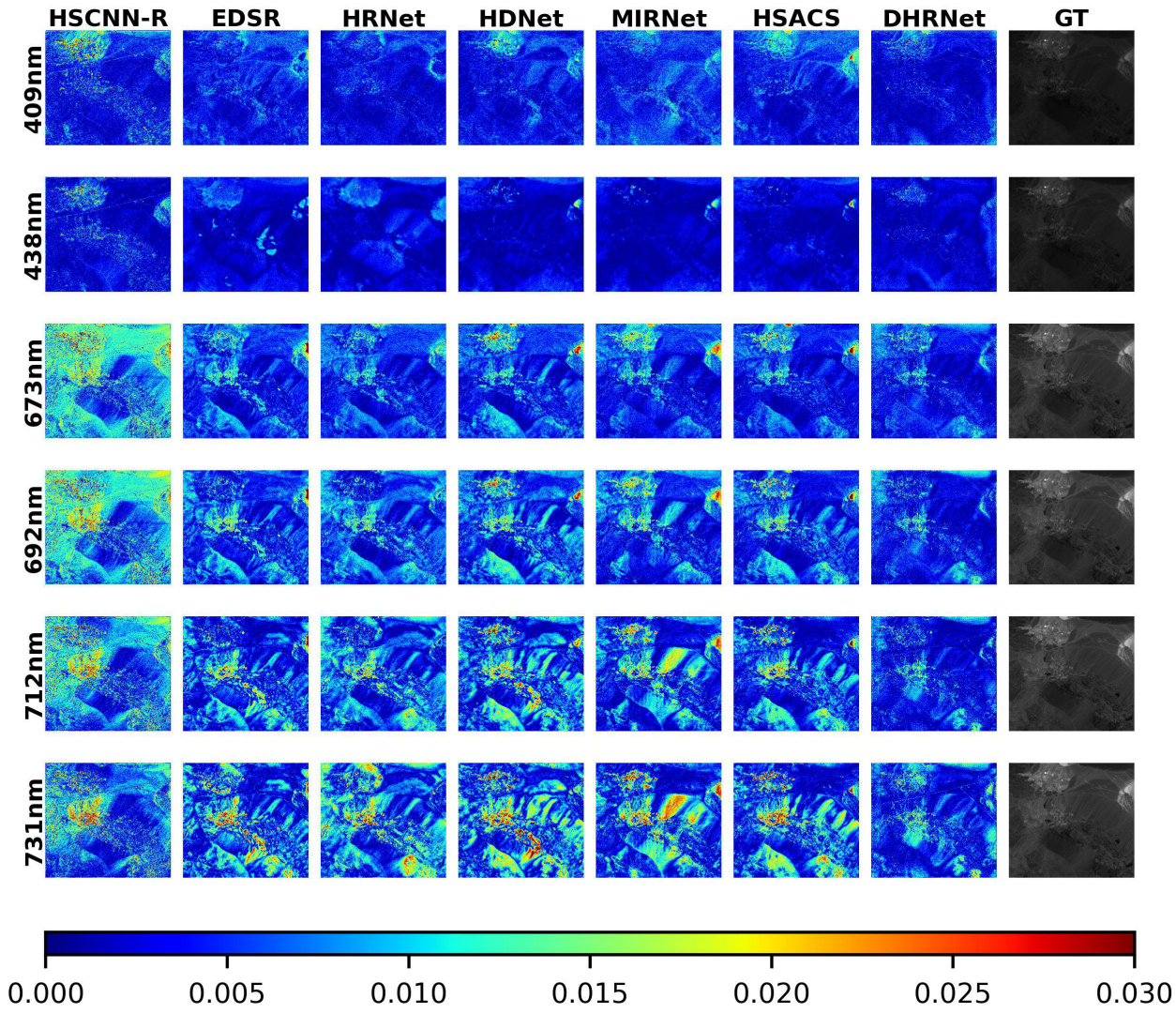


Fig. 5. Visual quality comparison of scenes from Cuprite dataset at six different bands.

TABLE II
QUANTITATIVE INDICATORS OF DIFFERENT METHODS ON THE NATURAL IMAGE DATASET

Datasets	Method	RMSE↓	PSNR↑	SSIM↑	SAM↓
CAVE	HSCNN-R [36]	0.0290	35.08	0.9710	9.71
	HRNet [37]	0.0228	36.04	0.9795	9.23
	EDSR [32]	0.0250	35.82	0.9805	9.34
	HDNet [30]	0.0248	36.96	0.9809	8.74
	MIRNet [50]	0.0223	36.33	0.9822	8.47
	HSACS [41]	0.0200	38.00	0.9841	7.24
	DHRNet	**0.0192**	**38.11**	**0.9849**	**6.99**

that the reconstructed results by DHRNet are approximately consistent with GT.

C. Results on Natural Image Datasets

Table II provides detailed quantitative comparisons of different methods on the CAVE dataset, from which it can be observed that the proposed DHRNet outperforms the introduced SOTA methods. Obviously, compared with the second-best method HSACS, DHRNet reduces the RMSE by 4% and the SAM by 3.45% while increasing the PSNR by 0.11 dB.

D. Ablation Study

In this section, ablation study is conducted to verify the effectiveness of each component in the proposed DHRNet, including the proposed SFEM, RFEM, multiscale learning strategy denoted as multi, and FEM. Table III provides the explicit quantitative results of DHRNet with or without the corresponding components on the GF5 dataset, which are denoted by symbols "✓" and "✘." Besides, $\boldsymbol{M}_b$, $\boldsymbol{M}_c$, and $\boldsymbol{M}_d$ present the models with each ingredient to be studied. Notably, $\boldsymbol{M}_a$ can be treated as the backbone with a similar structure as DHRNet, which processes the input shading feature and reflectance feature through a simple dual-pathway network consisting of residual stacking 2-D CNNs without any the abovementioned components.

1) Effect of SFEM and RFEM: As illustrated in Table III, there is a significant amelioration in both indicators after replacing 2D-CNNs in the $\boldsymbol{M}_a$ method with SFEM and RFEM, in which SAM achieves 10.84% decrease and RMSE achieves

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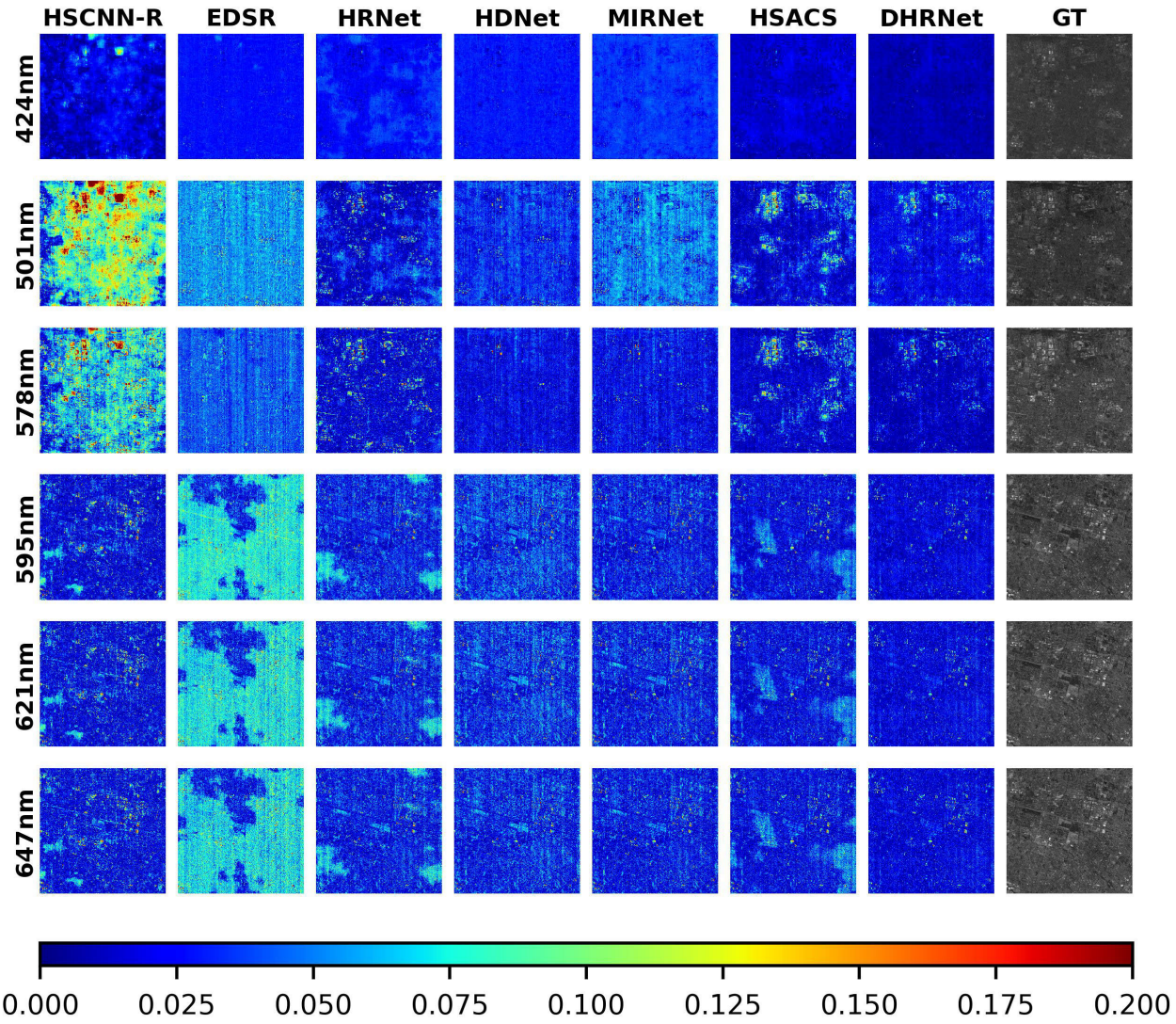


Fig. 6. Visual quality comparison of scenes from GF5 dataset at six different bands. Note that the displayed error maps are at size 512 × 512, which are cropped from the original validation sets due to the memory limitation.

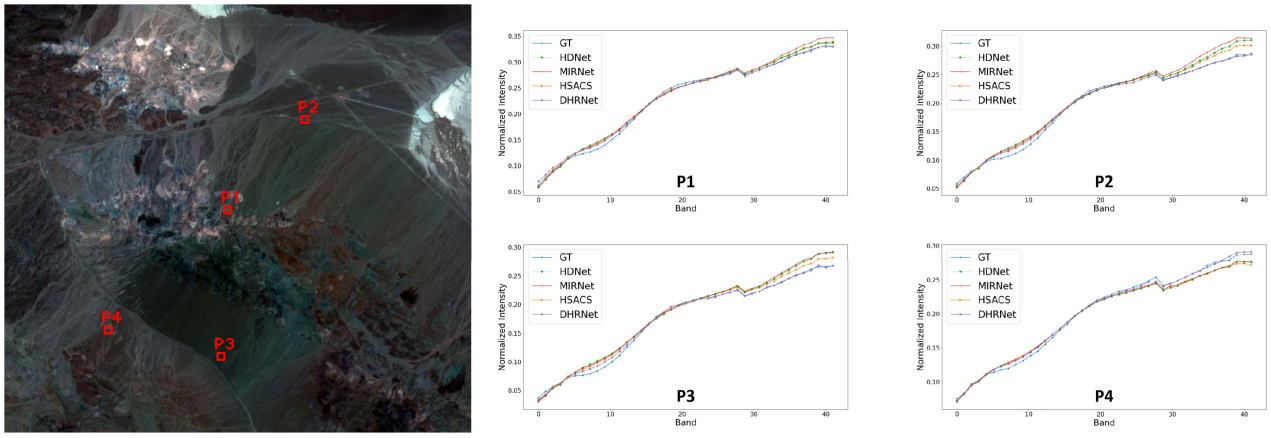


Fig. 7. Spectral signature analysis on Cuprite dataset. Note that "P1" to "P4" are four randomly selected points, which are highlighted on the left scene. The right four comparisons are the corresponding spectral curves of "P1" to "P4," respectively.

15.92% decrease. Moreover, compared with $\boldsymbol{M}_a$, $\boldsymbol{M}_b$ also achieves an apparent improvement in PSNR and SSIM, which are increased by 1.78 dB and 6.61%, respectively. It can be mainly attributed to the attention mechanism and dense structure embedded in the RFEM and SFEM. The RFEM can not only extract the shading image feature information

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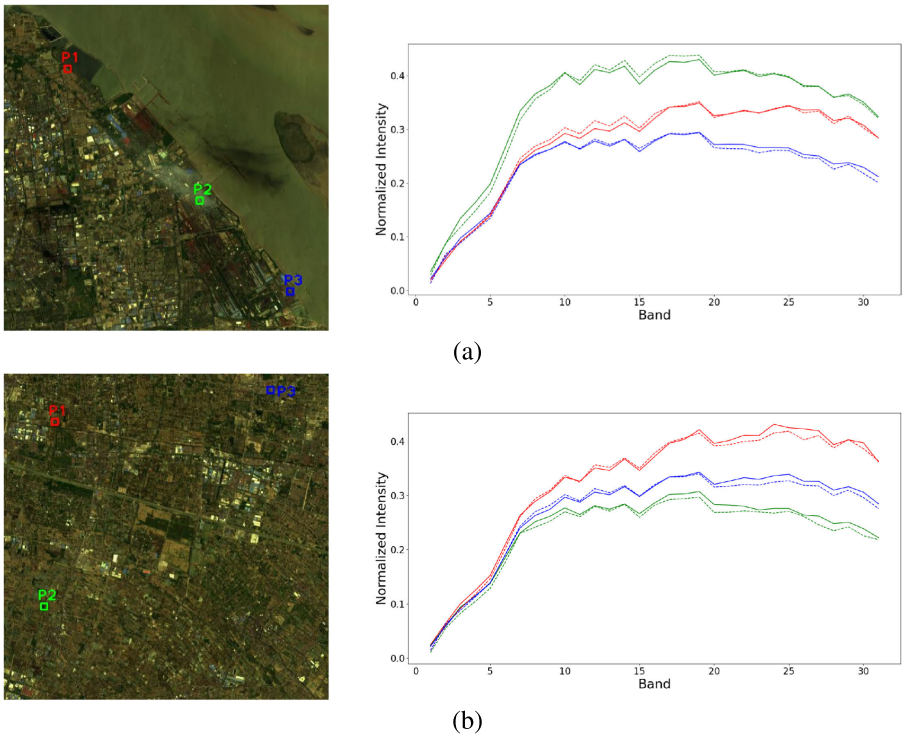


Fig. 8. Spectral signature analysis on the GF5 dataset. (a) and (b) Spectral curves from the GF5 dataset, where solid and dashed lines denote the GT and recovered spectrum, respectively.

TABLE III
ABLATION STUDY ON THE GF5 DATASET

Description	SFEM	RFEM	Multi	FEM	RMSE↓	PSNR↓	SSIM↑	SAM↓	Param(M)↓
M_a	✘	✘	✘	✘	0.0534	25.87	0.8566	3.23	**0.28**
M_b	✓	✓	✘	✘	0.0449	27.65	0.9132	2.88	0.42
M_c	✓	✓	✓	✘	0.0398	29.19	0.9221	2.79	5.84
M_d	✓	✓	✓	✓	**0.0383**	**29.84**	**0.9395**	**2.48**	5.90

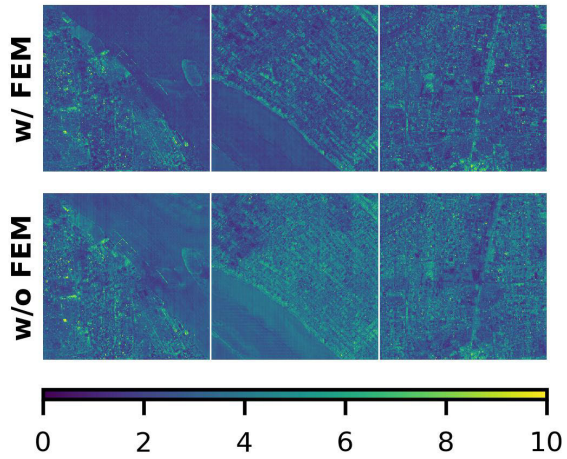


Fig. 9. Visualization of SAM from the GF5 dataset.

at different scales, but also utilizes attention mechanism to fully focus on the more meaningful information of spatial and spectral space, so as to improve the effect of reconstruction. The SFEM enhances the extraction of contextual feature information contained in the reflectance image by adopting dense structure. Therefore, SFEM and RFEM can effectively strengthen the discriminative learning of two different features.

2) Effect of Multiscale Strategy: This session mainly verifies the contribution of multiscale learning strategy designed in the proposed DHRNet. From Table III, it can be observed that the inputs are processed in different resolutions through employing the multiscale learning strategy, leading to obvious decrease in RMSE and slight decrease in SAM compared with method $\boldsymbol{M}_b$. Besides, $\boldsymbol{M}_c$ increases the PSNR and SSIM by 1.54 dB and 0.0089. Multiscale learning strategy can extract multiscale features while taking into account both local and global information through the pyramid structure, which can effectively promote the follow-up learning of extracted features, so as to obtain a better reconstruction result.

3) Effect of FEM: FEM is designed and deployed to improve the coarse HSI estimation. According to the results in Table III, $\boldsymbol{M}_d$ achieves the best performance after the refinement of FEM, which obtains an obvious improvement in PSNR and SSIM. Furthermore, RMSE goes down by 3.91%, and 0.31 is reduced in SAM compared with framework $\boldsymbol{M}_c$. Moreover, to better display the effectiveness of FEM, SAM is further described in Fig. 9, where three scenes from GF5 dataset are selected to visualize the difference in spectral domain. Dramatically, the global color of $\boldsymbol{M}_d$ is darker than $\boldsymbol{M}_c$ in all displayed scenes, indicating the obvious enhancement of spectral features after introducing the proposed FEM.

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It can be attributed to the ability of the proposed RAMB and introduced 3D-CNN stacked in FEM, which are capable of readjusting the importance of the bandwise feature along the depth dimension under the guidance of pure reflectance feature while regularizing the spatial–spectral consistency.

V. Conclusion

In this article, a novel framework DHRNet based on IID is proposed for spectral super-resolution of RGB images. IID is first conducted to decouple RGB input into the distinguished reflectance and shading feature. Then, a DHRNet with specifically designed multiscale learning strategy as well as SFEM and RFEM is further presented to fully and contrapuntally extract crucial features, especially for the bandwise features and spatial contextual features. A subsequent feature enhancement module (FEM) combined with cross-attention mechanism and 3-D CNN is devised to achieve the spectral enhancement while constraining the spatial–spectral consistency in hyperspectral domain. Moreover, a hybrid loss combined with smooth l_1 loss, SAM, and gradient prior is employed to effectively reduce spectral distortion and spatial degeneration. Experimental results on RS and natural scene datasets illustrate the effectiveness and superiority of the proposed DHRNet compared with SOTA frameworks both quantitatively and qualitatively.

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